



Started on	Thursday, 6 August 2026, 10:25 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:38 AM
Time taken	12 mins 39 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d",&n);
6      int i=1,s=1,count=2; //initialising i and s
7      while(s<=n)
8      {
9          count++; //increment i and adding s
10         i++;
11         s+=i;
12         count+=2;
13     }
14     count++; //end of while loop
15     printf("%d",count);
16     return 0;
17 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Thursday, 6 August 2026, 10:38 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:04 AM
Time taken	25 mins 23 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("");
                printf("");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  int main()
4  {
5      int n;
6      scanf("%d", &n);
7
8      int count = 0;
9
10     if (n == 1)
11     {
12         count++;           // if condition
13     }
14     else
15     {
16         count++;           // if condition (false)
17
18         for (int i = 1; i <= n; i++)
19         {
20             count++;        // i <= n
21
22             for (int j = 1; j <= n; j++)
23             {
24                 count++;    // j <= n
25                 count++;    // break
26                 break;
27             }
28         }
```

```
29         count++;           // j++
30         count++;           // i++
31     }
32
33     count++;                 // final i <= n (false)
34 }
35
36 printf("%d", count);
37
38 return 0;
39 }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Thursday, 6 August 2026, 11:29 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:38 AM
Time taken	8 mins 37 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int count=0;
3  void Factor(int num)
4  {
5
6      for (int i = 1; i <= num; ++i)
7      {
8          count++;
9          if (num % i == 0)
10         {
11             count++;
12             //printf("%d ", i);
13         }
14         count++;
15     }
16     count++;
17 }
18 int main()
19 {
20     int num;
21     scanf("%d",&num);
22     Factor(num);
23     printf("%d",count);
24     return 0;
25 }
26
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓

	Input	Expected	Got	
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Thursday, 6 August 2026, 11:40 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:47 AM
Time taken	6 mins 51 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int count=0;
3  void function(int n)
4  {
5      int c= 0;
6      count++; //assign c=0
7      for(int i=n/2; i<n; i++)
8      {
9          count++; //i<n
10         for(int j=1; j<n; j = 2 * j)
11         {
12             count++; //j<n
13             for(int k=1; k<n; k = k * 2)
14             {
15                 count++; //k<n
16                 c++;
17                 count++; //c++
18             }
19             count++; //k fails
20         } count++; //j fails
21     } count++; //i fails
22 }
23 int main()
24 {
25     int n;
26     scanf("%d",&n);
27     function(n);
28     printf("%d",count);
29     return 0;
30 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



Started on	Wednesday, 16 September 2026, 2:52 PM
State	Finished
Completed on	Wednesday, 16 September 2026, 2:57 PM
Time taken	5 mins 23 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
}
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2
3  int count = 0;
4
5  void reverse(int n)
6  {
7      int rev = 0, remainder;
8
9      count++; // rev = 0
10
11     while(n != 0)
12     {
13         count++; // n != 0
14
15         remainder = n % 10;
16         count++; // remainder = n % 10
17
18         rev = rev * 10 + remainder;
19         count++; // rev = rev * 10 + remainder
20
21         n = n / 10;
22         count++; // n = n / 10
23     }
24
25     count++; // n != 0 fails
26 }
27
28 int main()
29 {
30     int n;
31
32     scanf("%d", &n);
33
34     reverse(n);
35
36     printf("%d", count+1);
```

```
37 |  
38 |     return 0;  
39 | }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

